

Response to Reviewers

Paper 82 — EchoMind: Secure Full-Duplex Retrieval-Augmented Voice and Document Intelligence for Enterprise Knowledge Work — QASC 2026, Full Research Papers track

We thank all three reviewers. The camera-ready revision addresses every weakness raised. The central change is that the paper is no longer results-free: a new Sect. 8, "Measured Results" (six subsections, Tables 6–10), reports an instrumented evaluation of the implemented system — latency decomposition, a dual-loop ablation, permission-isolation testing including a path-level audit, prompt-injection containment across four defense arms, and grounding fidelity — under a protocol with deterministic judges, frozen query sets, and Wilson confidence intervals. Sect. 7.1 now documents the deployed configuration exactly. Where measurement contradicted the design's own expectations, the text says so explicitly rather than repairing the claim. Below, each comment is quoted and answered with section pointers.

Reviewer #2

Comment 1

"The paper is a reference design and research agenda without implementation results, latency measurements, or benchmark scores to validate the proposed architecture."

Addressed by the new Sect. 8. Table 6 fills the previously empty measurement template with per-stage medians, p95, and p99 ($n = 30$ per cell): the permission predicate costs 0.08 ms inside a 3,313 ms median grounded answer; Table 7 reports scaling at concurrency 1/4/16. Table 8 reports the dual-loop ablation that tests the paper's central claim: the production loop reaches first audio in 636 ms median with 0/10 assertion-invariant violations, a single loop is 36% slower, and removing invariant I1 alone produces ungrounded assertions on 10/10 turns at equivalent latency. Table 9 reports permission isolation (0/50 cross-tenant leaks; 0/359 leaked candidates in a per-path audit; one residual timing channel measured and reported open). Table 10 reports injection containment (attack success 10.2% with no defense, 5.1% with the evidence envelope, false positives 0/25). Sect. 8.5 reports grounding (citation precision 0.979, abstention 78.3% on unanswerable queries, zero fabricated citations). Two measured findings contradict expectations stated in the submitted version — retrieval terms are 2.8–9.5% of grounded-answer latency rather than dominant (Sect. 8.1), and containment is real but silent, so the "reportable event" property is not yet implemented (Sect. 8.4) — and both are reported unchanged.

Comment 2

"Key claims rest on internal technical notes and a company benchmark that are not publicly accessible, which limits reproducibility and independent verification."

The measured study depends on neither non-public source, and Sect. 7.4 now enumerates the artifacts released with the paper: the frozen 92-query set with per-stratum gold labels and its SHA-256, the full configuration manifest (exact model identifiers, revisions, quantization, index and fusion parameters, decoding settings, corpus composition), per-stage span logs behind every latency cell, the 98-instance adversarial corpus with class labels, the deterministic scoring scripts for every judge, and raw per-probe outputs. Reference [1] is cited only for product deployment configuration and [15] only for an evaluation taxonomy the paper restates in full; no architectural, evaluative, or measured claim requires either.

Comment 3

"The proposed evaluation weighting scheme is asserted rather than justified through experiments or user studies."

The revision asserts no weighting anywhere. Sect. 9.1 specifies how weights are to be derived — pairwise AHP elicitation from stakeholder panels with a consistency-ratio check — and requires a sensitivity analysis reporting the critical perturbation δ^* at which system rankings reverse; Table 13 is explicitly labeled as illustrative priors to be replaced by elicitation. Consistently with that discipline, no composite score is computed anywhere in the paper: all measured results in Sect. 8 are published per-dimension, and Sect. 8.6 states plainly that the elicitation has not yet been run with real panels, because synthesizing panel judgments would fabricate the very numbers the procedure exists to derive.

Comment 4

"Novelty is mainly in the integration of existing components such as Whisper, pgvector, and RAG rather than in new methods."

Two answers, one factual and one empirical. Factually, the implemented system is not that stack: Sect. 7.1 documents a two-pass NVIDIA NeMo ASR arrangement (streaming partials plus a second-pass decoder), an embedded FAISS/BM25/SQLite substrate with the tenant predicate evaluated inside the candidate scan, and a mixture-of-experts answer model in NVFP4 quantization served by TensorRT-LLM on a 128 GB unified-memory edge node — the architecture constrains ordering, invariants, and labeling discipline, not vendors. Empirically, the revision tests whether the claimed delta — the invariants and their ordering, not the components — carries the safety properties. It does: with components and latency held fixed, removing invariant I1 alone produces confabulation on every measured turn (Sect. 8.2, Table 8); a prompt-only injection defense is indistinguishable from no defense while the structural envelope halves attack success (Sect. 8.4, Table 10); and a per-path audit of the permission-before-retrieval property found and fixed an implementation-level violation that neither the architectural statement nor answer-level testing detected (Sect. 8.3). These are results about the arrangement, which is precisely the contribution claimed in Sect. 1.2.

Reviewer #4

Comment

"The manuscript is primarily conceptual and lacks experimental validation... Strengthening the empirical evaluation, clarifying the novel research contribution beyond existing RAG architectures, and reducing descriptive content would considerably improve the manuscript."

Empirical evaluation: the new Sect. 8 with Tables 6–10, as summarized under Reviewer #2, Comment 1. Contribution clarity: the abstract now leads with the measured results; each contribution in Sect. 1.2 carries a pointer to the measurement that tests it; and Sect. 12 states which claims the measurements settled and which remain open. Descriptive content: the architecture description that Reviewers #1 and #4 assessed positively is retained, while Sects. 1.1, 7, 11, and 12 were rewritten tighter; the net additions to the paper are results and their limitations, not description. The scope and validity limits of the study — single node, 421 content chunks, $n = 10$ ablation turns, a rule-based invariant judge, no tool layer — are stated in Sects. 8.6 and 11 rather than left for readers to discover.

Formatting

"Also, please follow the springer publishing guidelines and format the paper for final camera ready mentioned on the website."

The camera-ready uses the Springer single-column layout on A4 with 10 pt body text and numbered citations in Springer style; Fig. 1 was regenerated at high resolution and updated to name the two loops of the execution model; all tables carry captions; the acknowledgments placeholder was removed; and references [21]–[25] were added for components of the measured stack (Faiss, NeMo, BM25, Qwen3, Nomic Embed).

Reviewer #1

We thank the reviewer for the positive assessment and for recommending acceptance.